

Guillem Masdemont Serra

MSc Student in Artificial Intelligence

CONTACT INFORMATION	Rome, Italy Sapienza University of Rome	guillem.masdemont01@estudiant.upf.edu guillemmasdemont.github.io
ABOUT ME	Recent graduate in Mathematics and Physics, now in the second year of the Erasmus Mundus Joint Master's in Artificial Intelligence at Sapienza University of Rome under a full tuition fee waiver scholarship. Interested in research with strong mathematics, such as reinforcement learning and fluid mechanics.	
EDUCATION	MSc, Artificial Intelligence , Erasmus Mundus Joint Master (EMAI) 2025–Present Research-oriented joint program at Sapienza University of Rome (Italy), University of Ljubljana (Slovenia) and Universitat Pompeu Fabra (Spain). Acceptance rate below 2% Supported by an Erasmus Mundus full tuition-fee waiver scholarship Current average grade: 9.6/10. Expected graduation: August 2027 BSc, Mathematics and Physics , Autonomous University of Barcelona 2020–2025 Degree with the highest admission cut-off mark in the Catalan university system (2020) Final joint average grade: 8.94/10. Ranked 4th in the cohort. Erasmus exchange at University of Vienna (1st semester, 2024–25): 1.0/1.0 average in courses from the MSc in Advanced Mathematics Mathematics BSc. Thesis: <i>Application of a Spectral Method to Two-Dimensional Incompressible Flows in a Rectangular Domain</i> . Supervisor: Prof. Jezabel Curbelo (UPC). Awarded highest honors Physics BSc. Thesis: <i>Far-field heat transfer and persistent heat currents arising in a cylindrical non-reciprocal configuration</i> . Supervisors: Prof. Georgia Papadakis and Dr. Julien Legendre at ICFO. Led to this publication in <i>Physical Review Research</i>	
RESEARCH EXPERIENCE	Summer Research Fellow , ICFO – The Institute of Photonic Sciences Jul–Sep 2024 Thermal Photonics group of Prof. Georgia Papadakis, supported by La Pedrera Foundation fellowship Studied the breakdown of Kirchhoff's law in non-reciprocal systems for directional heat-flow control and implemented a computational framework to achieve heat rectification. This establishes benchmarks for perfect heat rectification. Barcelona Introduction to Mathematical Research , CRM/BGSmath Jul–Sep 2023 Dynamical Systems group, University of Barcelona (Profs. Núria Fagella and Xavier Jarque) Studied the topology of the escaping set for transcendental entire functions, and delivered a talk proving the nonemptiness of these sets (A.E.Eremenko, 1989).	
PUBLICATIONS	Far-field heat transfer and monochromatic thermal currents in a cylindrical nonreciprocal cavity 2026 G. Masdemont , J. Legendre and G. T. Papadakis. <i>Physical Review Research</i> 8 , 023293 (2026). (DOI: 10.1103/5bbx-qx7v)	

EMPLOYMENT	Private Mathematics Tutor, Freelance Current Teaching mathematics and physics courses at every level, from middle school and high school up to university.
HONORS AND AWARDS	Erasmus Mundus Scholarship, full tuition-fee waiver awarded by the European Union and the EMAI consortium 2025 ECIU Creathon Grant, represented UAB in Linköping, Sweden 2024 Baccalaureate Studies, Lacetània High School. 1st in the 2018–2020 cohort. University entrance exam mark: 13.51/14. 2020 Cangur Mathematics, 2nd place in central Catalonia 2019
OUTSIDE ACADEMIA	Violist, Camerata Bacasis Orchestra, semi-professional contract Current Class Delegate, Mathematics and Physics double degree, UAB 2023–2024 GLEE Mission, Colorado Space Grant Consortium 2022–2024 Member of the UAB student team in the Great Lunar Expedition for Everyone, developing and testing LunaSats for experiments on the lunar surface.
SKILLS	Coding and Software: Python (pandas, scikit-learn, Hugging Face Transformers, Matplotlib, Gymnasium...), C, R, SQL, Git, $\text{\LaTeX}$ Languages: Catalan and Spanish (native), English (C1, Cambridge Advanced score 192), Italian (currently learning) Interests: Hiking and cooking